NAMAN YESHWANTH KUMAR

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SUMMARY

MS Computer Engineering (ASU, GPA 3.81, May 2026) building AI-driven software systems for VLSI circuit design — agentic workflows and autonomous optimization loops for end-to-end design automation. Built SiliconCrew, a self-improving LangGraph agent that drives RTL-to-GDSII through synthesis, place-and-route, and sign-off — scoring **37/92 on CVDP** and **5/5 on ASU Spec2Tapeout** with zero human intervention. Drove a GNN ASIC end-to-end on ASAP7 7nm through Synopsys DC, Cadence Innovus, and Voltus.

TECHNICAL SKILLS

AI for EDA & Circuit Design: LangGraph | LangChain | ReAct Agents | MCP Protocol | Multi-provider LLM Routing | PyTorch | TensorFlow | CUDA (hand-written kernels) | Python | C++ | FastAPI

Synthesis, PD & Sign-off: Synopsys Design Compiler | Cadence Innovus | Voltus | Calibre DRC/LVS | OpenROAD | Yosys | SymbiYosys | Static Timing Analysis | NLDM Liberty | SDC | RTL-vs-Netlist Equivalence Checking

RTL, Verification & VLSI Process: SystemVerilog | Verilog | VLSI Design | Cocotb | iverilog | gem5 (C++ internals) | ASAP7 7nm FinFET | SkyWater 130nm | Docker | Linux | Git

EDUCATION

Master of Science — Computer Engineering (GPA 3.81/4.0) **May 2026**
Arizona State University **Tempe, AZ**
Coursework: VLSI Design · VLSI Design Automation · Advanced Computer Architecture · Algorithm/Hardware Co-Design · SoC Design

Bachelor of Engineering — Electrical and Electronics **August 2022**
BMS College of Engineering **Bengaluru, India**

PROJECT EXPERIENCE

SiliconCrew — Self-Improving Agentic AI System for End-to-End Circuit Design **Nov 2025 – Present**

- Built SiliconCrew, an autonomous LangGraph ReAct agent (21 tools) that drives the full circuit-design pipeline — RTL generation → lint → simulation → Yosys synthesis → OpenROAD place-and-route → post-synthesis simulation → GDSII — across SkyWater 130nm and ASAP7 7nm PDKs, with SymbiYosys bounded model checking and Cocotb constrained-random; FastAPI + WebSocket backend, full MCP server (stdio/SSE/HTTP) compatible with Claude Desktop, VS Code, and any MCP client.
- Implements an autonomous optimization loop: on simulation failure the agent reads VCD waveforms, diagnoses signal-level bugs, patches RTL, and re-verifies — system prompt mandates "never guess, always read waveforms" and a minimum of 3 improvement iterations until QOR targets are met; enforces SDC constraint validation, RTL-vs-netlist equivalence checking, and signoff guardrails before tape-out.
- Achieved **37/92 first-pass on CVDP** (Comprehensive Verilog Design Problems) and **5/5 on ASU Spec2Tapeout** with zero human intervention; multi-provider LLM routing across Gemini, OpenAI, and Anthropic.

Graph Neural Network ASIC Accelerator — Full RTL-to-GDSII on ASAP7 7nm **Spring 2025**

- Designed an 11-module GNN ASIC accelerator in SystemVerilog — 8 parallel MACs for 96-element dot products and a 12-state FSM for bidirectional COO edge aggregation — and completed the full RTL-to-GDSII flow: Synopsys Design Compiler synthesis on ASAP7 7nm RVT (1 ns clock, 10,878 cells), Cadence Innovus APR with CTS, and Voltus post-APR power analysis driven by VCD activity files.
- Achieved **50,300 µm²** chip area at 50.6% core density, **52.29 mW** total power (64.8% switching), and **99.2 ns** end-to-end latency.

Static Timing Analysis Engine — Python (Sign-off-grade) **Fall 2025 – Spring 2026**

- Built a complete Static Timing Analysis engine in Python from scratch — Liberty NLDM parser (7×7 delay/slew LUTs with bilinear interpolation), .bench circuit-graph parser, topological sort via Kahn's algorithm, forward arrival-time and slew propagation, backward RAT/slack computation, and critical path extraction; validated against ISCAS benchmarks c17, b15, and c7552.

CUDA MLP — Hand-Written GPU Kernels for Neural Network Training **Fall 2024**

- Implemented 6 CUDA kernels in C++ by hand for end-to-end MLP training on MNIST — linear forward (matmul + bias), linear backward (gradients w.r.t. inputs/weights/biases), and ReLU forward/backward; configured block/thread launch dimensions per kernel, managed host↔device transfers explicitly with manual cudaMalloc/cudaMemcpy, and bound the kernels to a Python/PyTorch training loop through ctypes FFI.

LRU-IPV Cache Replacement Policy — gem5 C++ Internals **Fall 2024**

- Implemented LRU with Insertion Position Vector (IPV) replacement policy from scratch inside gem5's C++ source as a first-class SimObject, integrating all 4 cache interface methods (reset, touch, getVictim, invalidate); modified 5 gem5 source files and validated across 3 L2 cache configurations (128/256/512 kB), observing L2 replacements collapse from **637 → 112 → 2** as cache capacity grew — isolating capacity-vs-policy effects at 41.8M instructions, 2.04 IPC.

WORK EXPERIENCE

Systems Engineer **Oct 2022 – Apr 2024**
NCR GOLD **Bengaluru, India**

- Automated the entire order lifecycle for a wholesale business — production Django platform (PostgreSQL + Celery + Nginx) processing **18,000+ orders over 3 years**; reverse-engineered third-party ERP (zero public API) via SQL schema analysis for bidirectional sync, eliminating ~20 min of manual relay per order.

AWARDS & HONORS

- Won **'Best Use of AI'** at Strategy X DevHacks'25 (ASU) — built AdaptED AI, a Gemini-powered platform that generates personalized flashcards, quizzes, and learning paths from uploaded documents in a 24-hour hackathon.
- Ranked **Top 10 nationally** in e-Yantra Robotics Competition (IIT Bombay) — designed a self-balancing vehicle in CoppeliaSim, February 2022.
- Secured **2nd place at Hack SoDA 2024** (Amazon-sponsored, 24-hour) — built PassGen, a secure offline Chrome Extension for unique password generation.